Smart Crime Insights: Patterns and Demographics Analysis in Los Angeles City

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Introduction

What's the Problem?

- Uncover temporal, spatial, and demographic patterns in Los Angeles city crime data;
- Assess how the COVID-19 pandemic influenced crime dynamics;
- Examine relationships between crime trends and socio-economic factors to forecast future patterns;
- Generate an interactive dashboard to visualize these insights.

Why is it important and why should we care?

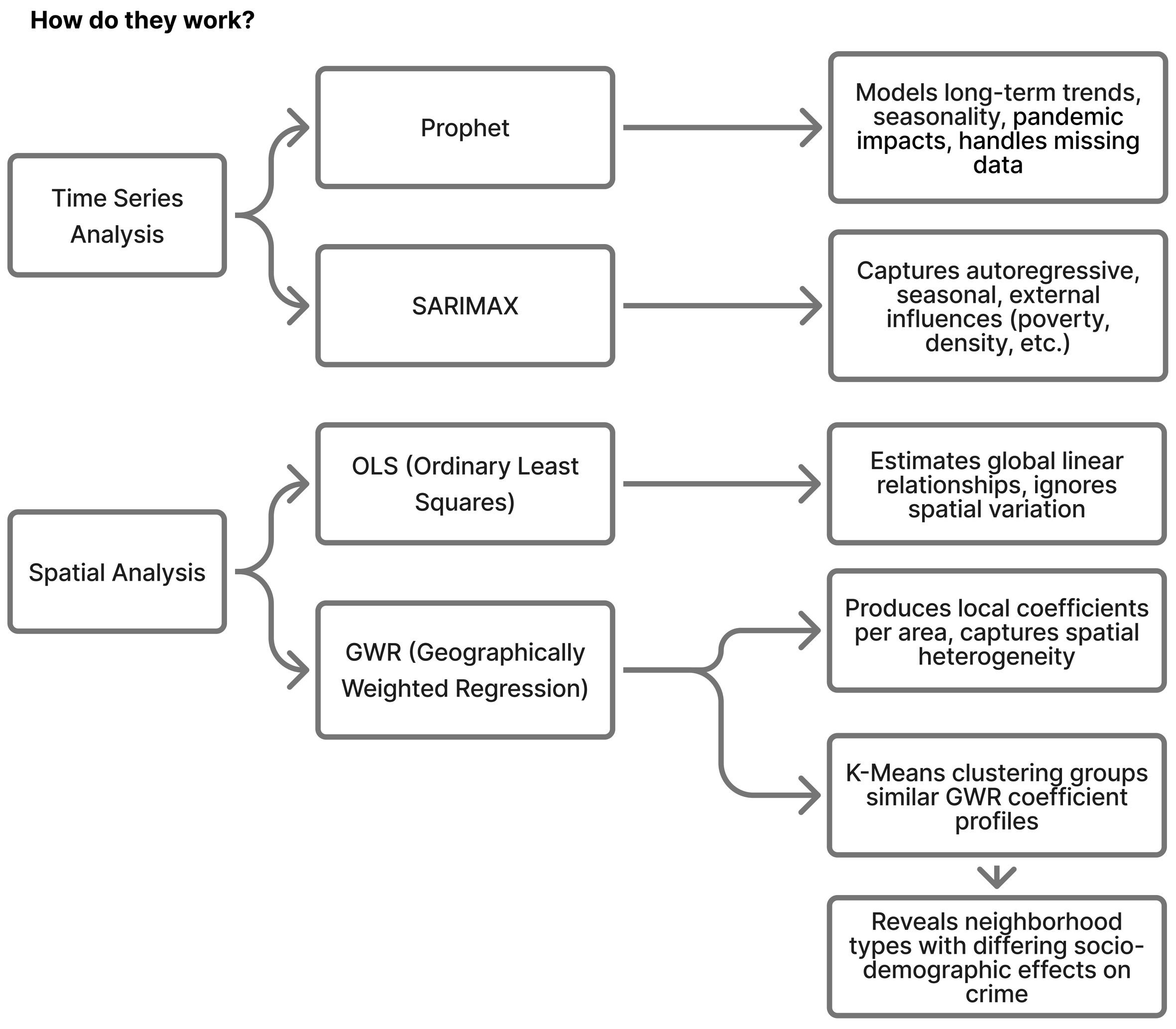
Crime is a key indicator of urban stability and public safety. Shifts in population, geography, and economic conditions, along with external events like the pandemic, can reshape crime patterns. Understanding these interactions helps reveal underlying drivers of crime and supports more effective prevention and intervention strategies.

Our Approaches

What are they?

- Dual-Model Comparative Framework to do time series analysis
- OLS and Geographically weighted regression (GWR) to do spatial analysis

How do they work?



Why do they work?

- Crime has both predictable patterns and external influences → SARIMAX is for interpretable seasonal structures, enabling interpretation of socio-demographic effects; Prophet is for nonlinear trend detection, capturing pandemic impacts.
- Crime is spatially heterogeneous → OLS finds "global rules"; GWR finds "local rules"

What is new?

- Dual-Model Comparative Framework

Using SARIMAX and Prophet enables a robust comparison of linear and additive decomposition methods, **capturing both localized seasonality and long-term trends**.

- Pandemic-Aware Temporal Modeling

A COVID-19 dummy variable isolates short-term disruptions and reveals post-2022 recovery dynamics. Socioeconomic and Demographic Integration: Incorporating poverty rate, population density, and aging population data enhances explanatory power and policy relevance.

- GWR Model

As we analyzed the crime dataset across different zip code areas in Los Angeles, we noticed that the geographical variables might capture certain useful information for our models, which has been neglected. To **capture this spatial variations**, our new method focuses on local indicators of spacial-association.

- Interactive Visualization Dashboard

User-friendly and designed for users of all backgrounds to find what they need.

Data

How to get data: Manually Downloaded

Manually download all the data, then integrate Los Angeles Police Department (LAPD) crime data (2013–2023) with U.S. Census demographic indicators at the ZIP Code Tabulation Area (ZCTA) level.

What are the characteristics:

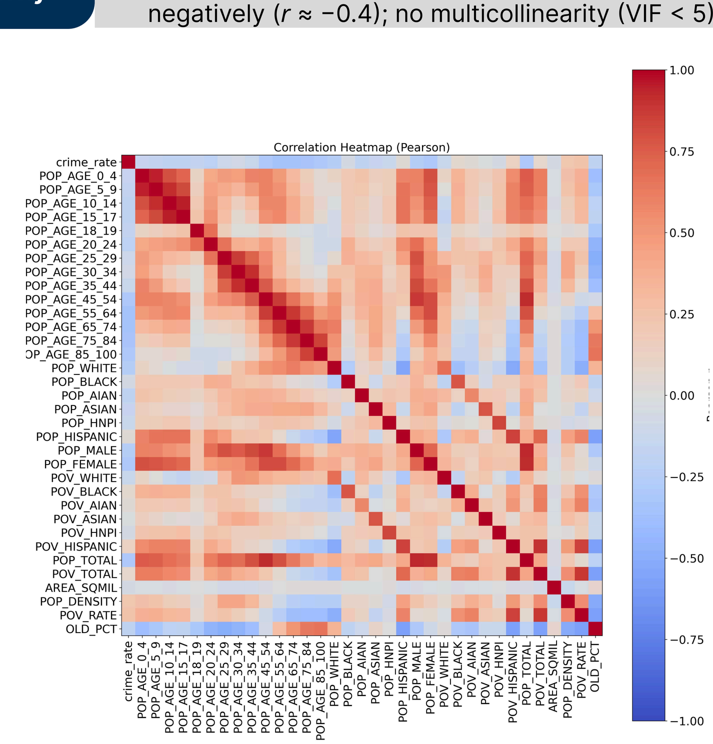
- 2,398,393 crime records (656.6 MB on disk)
- 30,804 census tract records (8.6 MB on disk)
- The crime dataset includes detailed information such as crime type, reported date and time, and victim characteristics. The census dataset provides demographic and socioeconomic indicators for each geographic area, including population distribution and poverty statistics at the tract level.

Experiments and results

How was the project evaluated?

Experiment 1 Variable Sensitivity

- Correlation,VIF tests (VIF < 5), Random Forest Feature importance;
- Poverty and density correlate positively with crime ($r \approx +0.5$), elderly share negatively ($r \approx -0.4$); no multicollinearity (VIF < 5)



Experiment 2 Model Benchmarking

- Prophet with a pandemic variable has the lowest RMSE;
- SARIMAX achieves better AIC/BIC balance and local stability.

Experiment 3 Pandemic Effect

Pandemic dummy improves short-term fit but overestimates post-2023; removing it aligns forecasts with LAPD's observed decline

Experiment 4 Spatial Consistency

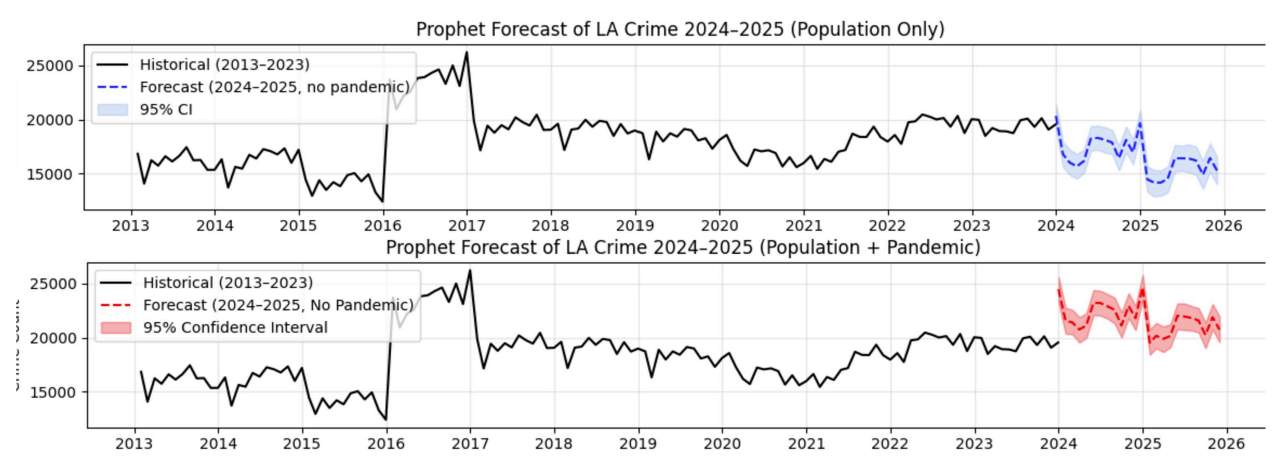
SARIMAX delivers smoother, interpretable seasonal cycles and lower residual variance, indicating stronger adaptability to local dynamics.

Experiment 5 Geographical Impacts

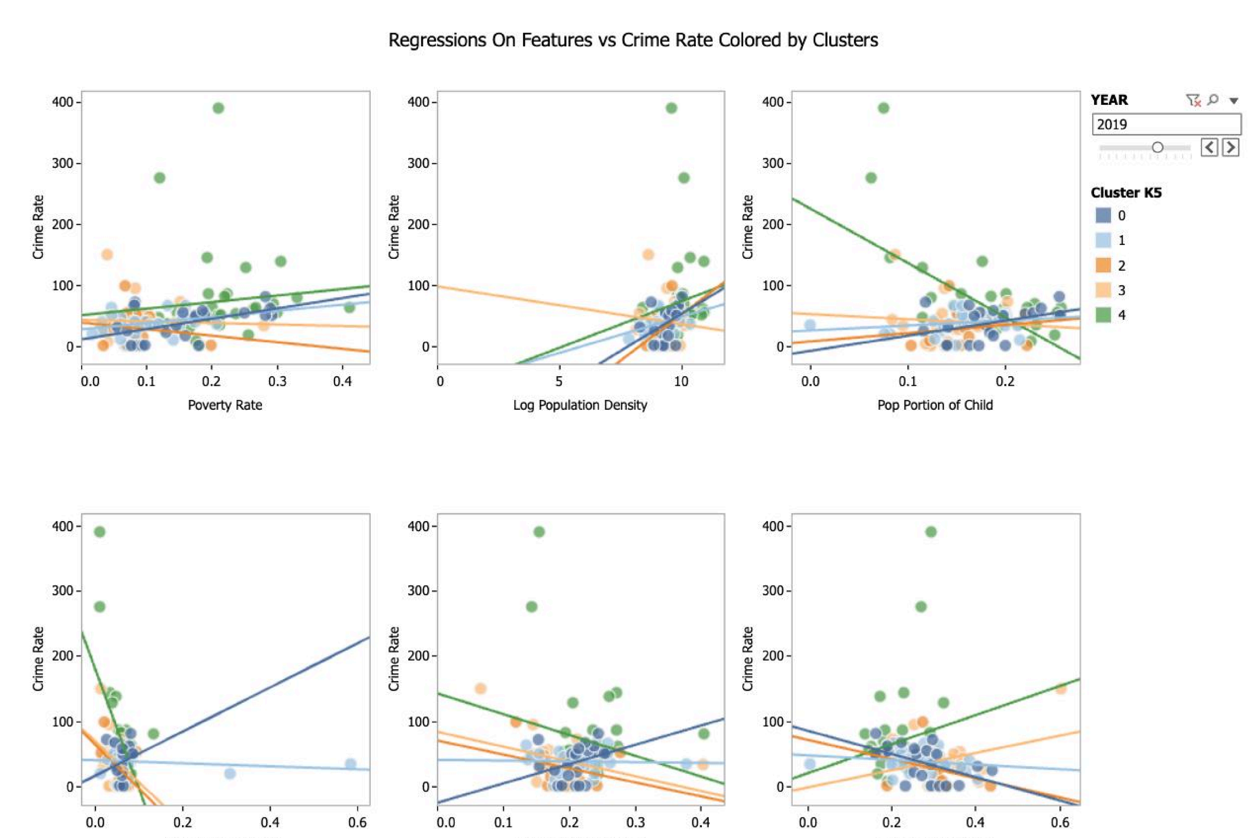
GWR clearly outperformed OLS by increasing R-squared and reducing spatial autocorrelation in residuals.

What are the results?

1. **Time series results:**
   - The COVID-19 pandemic caused short-term disruptions in Los Angeles crime patterns but no lasting structural change.
   - Prophet captured abrupt shifts during 2020–2022 and broader seasonal trends, while SARIMAX yielded higher neighborhood-level accuracy and produced more stable and interpretable long term forecasts.
   - Excluding the pandemic variable reduced Prophet's post-2023 overestimation, aligning with LAPD data showing a gradual decline.



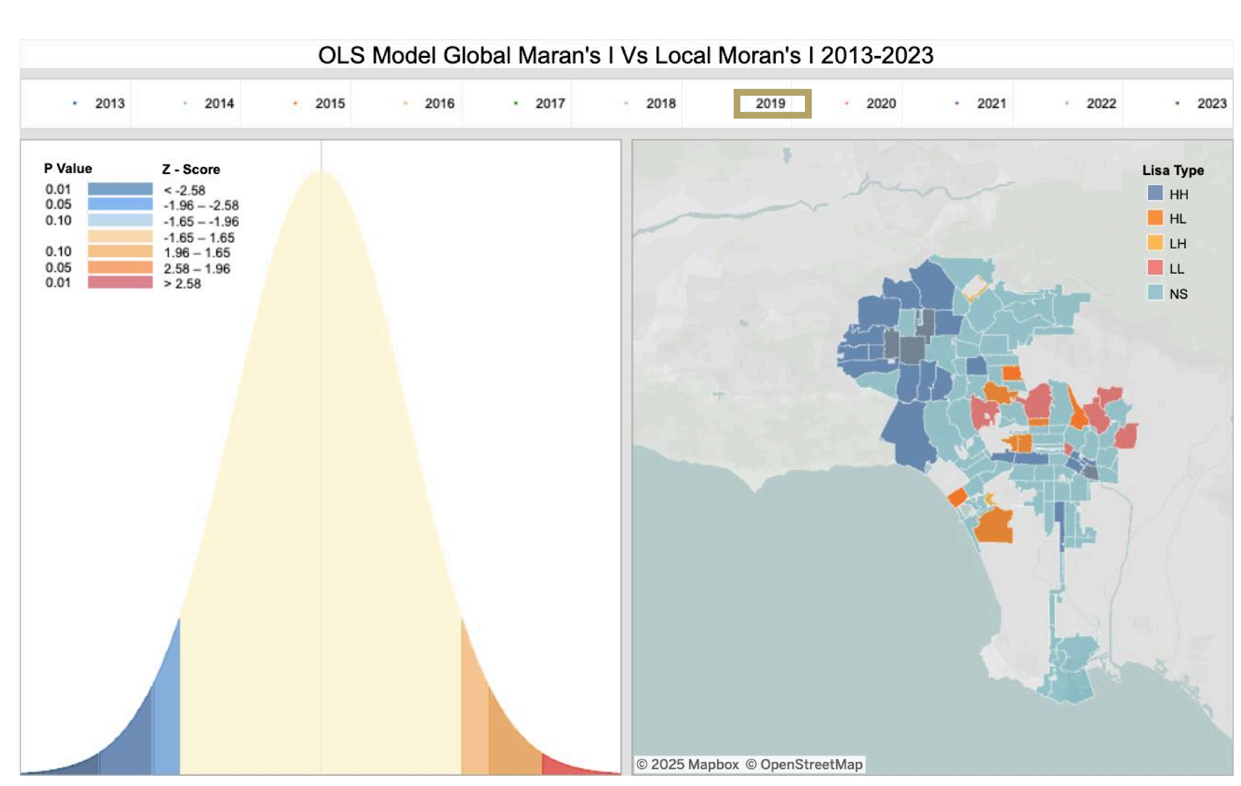
2. **Spatial analysis results:**
   - Crime drivers in Los Angeles vary significantly across neighborhoods and cannot be fully captured by global models.
   - While OLS identifies poverty as the strongest citywide predictor, GWR reveals substantial local heterogeneity, such as poverty, population density, and age composition affect crime differently across regions.
   - K-Means clustering further groups ZCTAs into distinct spatial regimes, highlighting meaningful patterns in how socioeconomic conditions shape crime.

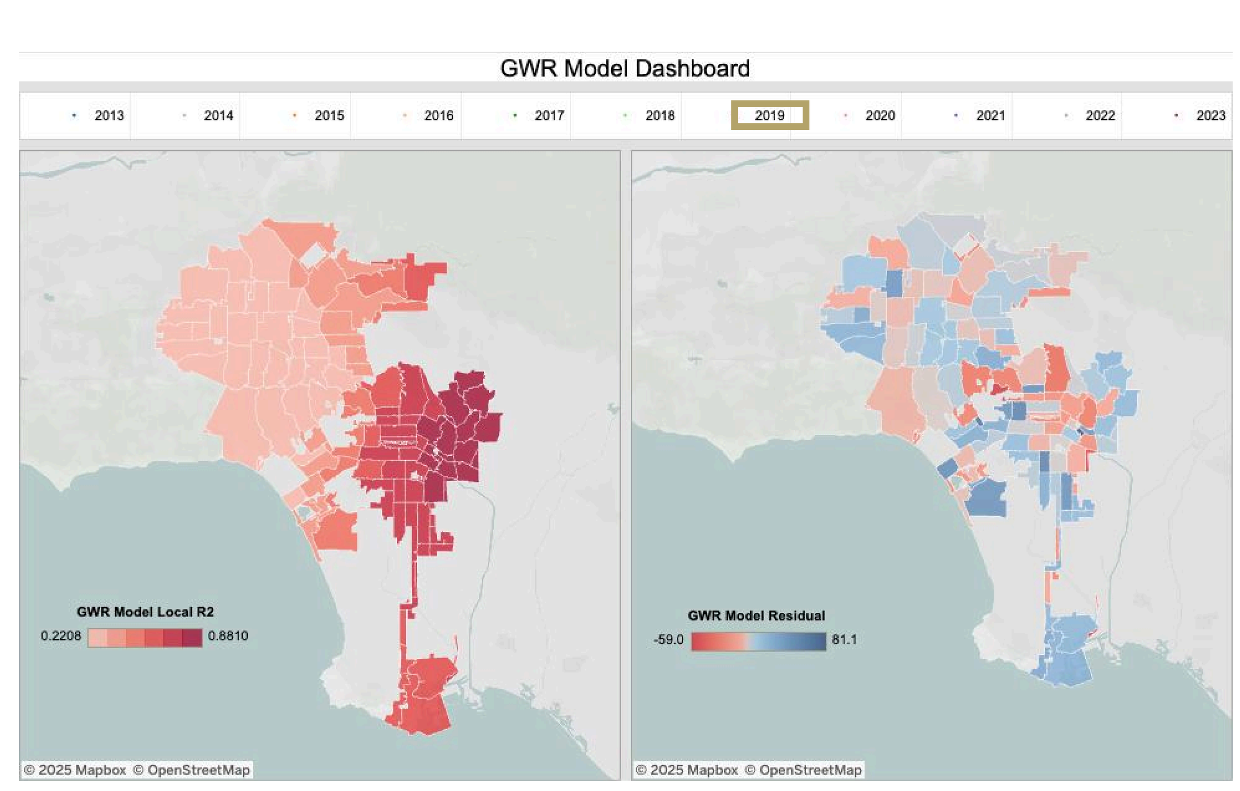


How do the methods compare to other methods?

1. Compared with ARIMA or ML models that focus only on temporal patterns, combining SARIMAX and Prophet improves interpretability and robustness, explains both how and why crime changes, enhances forecasting accuracy, and offers a more policy-relevant framework for post-pandemic urban analysis.

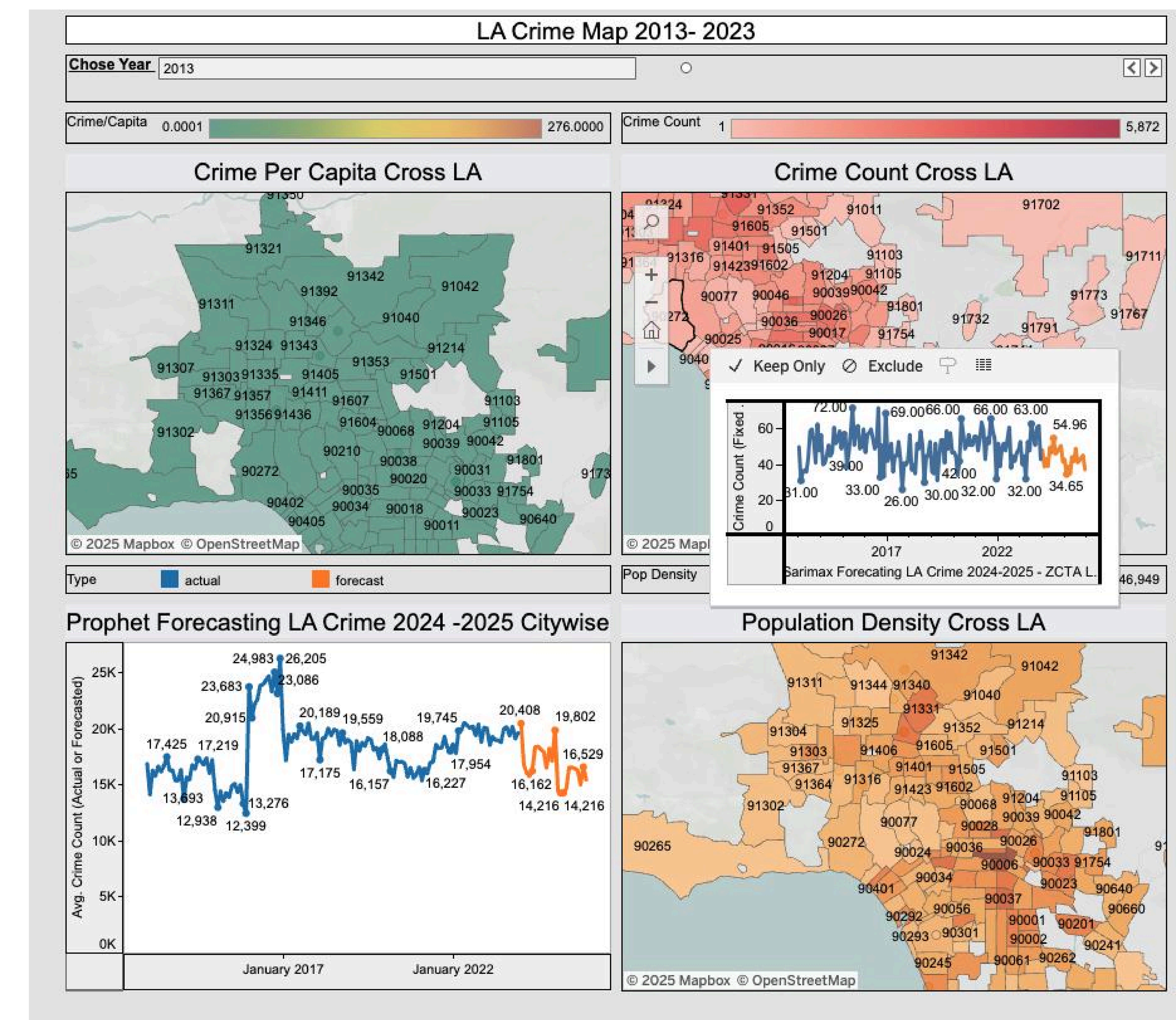
2. Compared with global OLS model, GWR effectively captures spatial heterogeneity and delivers substantially better performance. It also provides deeper, more actionable insights than citywide averages, underscoring the importance of place-specific analysis in crime research.
   - Taking 2019 for example, fit improves (R-square rises from 0.449 to 0.521), errors drop meaningfully (MAE 22.85→18.35; RMSE 33.81→25.71), and the information criterion favors GWR (AIC/AICc 1258.79→1237.17). Just as important, spatial autocorrelation in residuals is largely removed: OLS leaves clustered errors (Moran's I = 0.1795, p = 0.003), while GWR's residuals are effectively random (Moran's I = −0.037, p = 0.323).



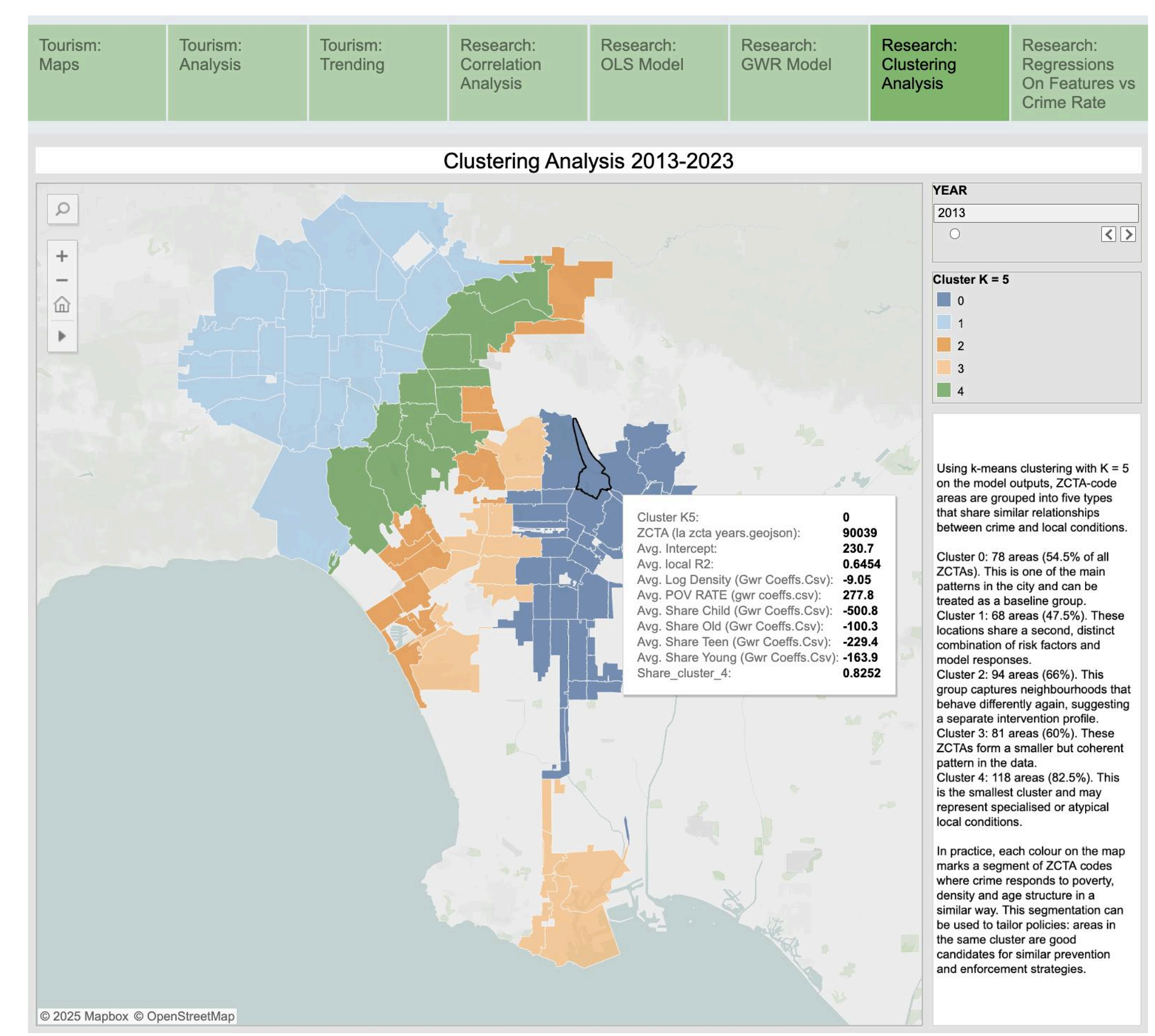


Interactive Visualization Effect

- Our interactive dashboards enable a clear, multi-layered exploration of Los Angeles crime patterns by combining spatial, temporal, and demographic views. The choropleth map displays crime at the ZCTA level, allowing users to hover over each ZIP code area to see total incidents, category-specific breakdowns, and year-by-year trends. This makes geographic disparities immediately visible—for example, central and south-central Los Angeles consistently show higher crime intensity compared with coastal and suburban neighborhoods.



- Complementing the map, our bubble and bar-based visualizations summarize where crimes occur across the city, highlighting that "Street," "Single-Family Dwelling," and "Multi-Unit Dwelling" remain the most common crime locations. Users can compare shifts in these patterns daily, hourly, or, before, during, and after COVID-19 to understand how mobility restrictions and reopening phases influenced crime distribution.



- Together, these interactive visual tools allow users to explore spatial hotspots, temporal fluctuations, socioeconomic context, and pandemic-related disruptions, making complex crime–census relationships both interpretable and actionable.